AI Secure Data Intelligence Platform

Real-Time Sensitive Data Detection System

Software Development

Student Details

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Technologies:	React, Spring Boot, MongoDB
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1. Introduction

This project presents an AI-powered platform designed to detect sensitive data leaks in real-time. The system analyzes multiple input types such as logs, text, SQL queries, and files to identify security risks and prevent data exposure.

2. Problem Statement

In modern applications, sensitive information like API keys, passwords, and personal data often gets exposed unintentionally through logs, code snippets, and database queries.

- Developers lack automated tools to detect such leaks
- Manual review is slow and error-prone
- Security breaches lead to financial and reputational loss
- SQL injection attacks remain a major threat

There is a need for a unified system that can automatically detect and classify such risks.

3. Solution

The AI Secure Data Intelligence Platform provides a real-time detection system that scans input data and identifies sensitive information using pattern recognition and intelligent analysis.

- Detects sensitive data using regex-based patterns
- Classifies risks into LOW, MEDIUM, HIGH, CRITICAL
- Applies policies like masking or blocking
- Generates human-readable insights
- Stores scan results for auditing

4. Key Features

- **Multi-Input Support**
  - Text, logs, SQL queries, and file uploads
- **Sensitive Data Detection**
  - API keys, passwords, emails, tokens
- **SQL Injection Detection**
  - Detects UNION, OR 1=1, DROP TABLE attacks
- **Risk Scoring System**
  - Assigns score from 0 to 100
- **Policy Enforcement**
  - Mask sensitive values or block high-risk input
- **AI Insights**
  - Provides explanations for detected risks

5. System Architecture

The system follows a layered architecture:

- Input Layer – Accepts user input
- Processing Layer – Extracts and prepares data
- Detection Engine – Identifies sensitive patterns
- Risk Engine – Calculates risk score
- Policy Engine – Applies actions
- Database Layer – Stores results

6. Working of the Project

The working of the system can be explained in the following steps:

1. User provides input (text, file, or SQL query)
2. System processes input using backend services
3. Detection engine scans for sensitive data
4. Risk engine assigns a risk score
5. Policy engine applies masking or blocking
6. Results are displayed on the frontend
7. Data is stored in MongoDB for history tracking

7. Demonstration Example

Example input:

```
API_KEY=sk-abc123
password=Admin@123
SELECT * FROM users WHERE id=1 OR 1=1;
```

System output:

- Detects API key and password

- Identifies SQL injection
- Assigns high risk score
- Masks sensitive data

8. Technologies Used

- Frontend: React, Vite, Bootstrap
- Backend: Spring Boot (Java 17)
- Database: MongoDB Atlas
- File Processing: Apache Tika
- Deployment: Netlify and Render

9. Advantages

- Real-time detection
- Automated security analysis
- Supports multiple data formats
- Reduces manual effort
- Improves system security

10. Conclusion

The AI Secure Data Intelligence Platform provides an effective solution for detecting and preventing sensitive data leaks. It combines intelligent detection, risk analysis, and policy enforcement to enhance application security.